04 — Technical Architecture Summary

What's actually live — written for a grant reviewer

> 💡 **KEY TAKEAWAY**
>
> Every capability below is live in the production codebase today — a cryptographic hash-chained audit ledger, GS1 Digital Link QR, a multi-role RBAC trust workflow, and PPWR-aware ESG analytics — not a roadmap promise.

Next.js (App Router) · TypeScript · PostgreSQL — Neon · Prisma ORM · Prisma.Decimal precision · SHA-256 hash-chained ledger · GS1 Digital Link · @react-pdf/renderer

A grant reviewer evaluating a digitalization or circular-economy project typically needs to answer two questions beyond "does the idea make sense": is there a real, working system behind this application, and is the technical approach sound enough to scale and survive regulatory/security scrutiny. This section answers both directly — the platform is not a prototype, it is a deployed application with a live PostgreSQL database, real API routes, and multiple working user-facing portals.

The Digital Product Passport data model

At the center of the schema is `PolymerPassport` — one row per physical batch — holding GTIN, batch ID, polymer type and grade, ISO 1133 melt flow index, density, tensile strength, recycled content (PCR/PIR distinction), bio-based content, and a verified carbon footprint figure, optionally third-party-verified. Every passport carries a SHA-256 `recordHash` and, once revised, a `previousRecordHash`, so the record's own history is independently checkable.

Built on top of this core, already live: **batch genealogy** (a recursive upstream traceability tree, capped at a bounded depth for query safety); a **carbon footprint calculator** kept strictly separate from the manufacturer's own verified LCA figure and always labeled illustrative; a **certification workflow** (ISCC PLUS, ISO 14021, EN 15343, REDcert2) with a formal pending-review → verified/rejected lifecycle, with expiry computed at read time rather than stored as a manually maintained flag; and a real **GS1 Digital Link QR** resolver with a printable label-sheet suite for factory-line batch labeling.

Cryptographic immutability and the audit ledger

Every consequential state change is written to an append-only `AuditLog` inside the same database transaction as the change itself. Each entry is SHA-256 hashed over its own fields plus the previous entry's hash, forming one global, insertion-ordered hash chain — the same structural principle blockchains use for tamper-evidence, without distributed-ledger overhead. Two independent layers of protection follow: a live integrity-verification function that mathematically proves whether any row has been altered, and a database-level append-only trigger that rejects any `UPDATE`/`DELETE` outright. A dedicated Audit Log interface lets an administrator filter, page, and export the full chronological trail, including a compliance-export workbook that stamps a live integrity-check result into the file at export time.

Role-based access and multi-party trust

A genuine multi-party trust model, not a single-tenant admin tool: **manufacturers** create and manage batch passports; **independent auditors** each hold their own individually issued cryptographic credential — not a shared role key, because a formal sign-off is only meaningful if attributable to one accountable person — and review submissions through a dedicated Auditor Portal; **B2B buyers** access their purchased-batch portfolio through a versioned Buyer API and read-only Customer Portal; **upstream suppliers** can be invited via a token-based, self-service submission flow without requiring full accounts; and **consumers** scanning the public QR code see only verified, publicly appropriate data — enforced server-side, not just in the UI.

ESG, compliance, and analytics layer

Computed fresh on every request rather than cached, so it can never disagree with the underlying records: PCR/PIR mass tracking and illustrative EU plastic-tax exposure; a certification compliance breakdown with live-computed expiry alerting; a transparent, weighted Sustainability Readiness Index; PPWR 2030/2040 recycled-content target tracking benchmarked against the regulation's published Article 7 thresholds; and a supplier performance/compliance ranking dashboard. Every computed score shows its full component breakdown and weighting rather than a bare number.

🛠️ **PLANNED, NOT YET BUILT — STATED FOR TRANSPARENCY**

Production-grade session authentication for the manufacturer/admin console (current interim mechanism uses shared per-role API keys, adequate for a pilot); a first-class batch-weight field so mass-based ESG metrics cover every batch, not only those where the carbon calculator was run; and formal third-party security/penetration review ahead of handling external paying-customer data at scale.